大学化学期末复习配套考点题目（致新书院）

ZJJ（配套课件笔记使用）专一针对大学化学，化学原理的朋友也可快速扫扫

宗旨是帮大家取得一个理想的分数（期末未完待续）

期末不涉及期中前内容的考察

Ch 11

Point 1 分子间作用力及各种比较

3. In which of the following molecules is hydrogen bonding likely to be the most significant component of the total intermolecular forces?

A) CO_2 B) $C_6H_{13}NH_2$ C) CH_4 D) $C_5H_{11}OH$ E) CH_3OH

4. What types of intermolecular forces exist in a mixture of NH_3 and H_2S?

A) dispersion forces and hydrogen bonds
B) dispersion forces, dipole-dipole forces, and hydrogen bonds
C) dispersion forces, hydrogen bonds, and ion-dipole forces
D) dispersion forces
E) dispersion forces and dipole-dipole forces

注意氢键对应元素有且只有那三个

5. ________ are particularly polarizable.

A) Small nonpolar molecules
B) Small polar molecules
C) Large nonpolar molecules
D) Large polar molecules
E) Large molecules, regardless of their polarity

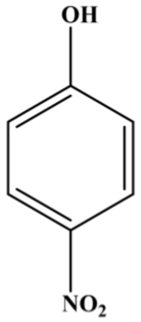
6. Of the following, ________ is the most volatile.

A) CH_2Br_2 B) CH_4 C) CH_3Br D) $CHBr_3$ E) CBr_4

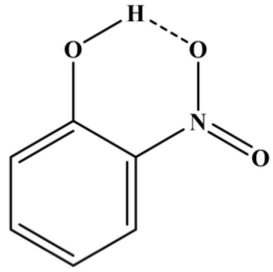
7. Which of the following is the correct order of boiling points for $BaCl_2$, H_2, CO, HF and Ne?

A) $H_2 < Ne < HF < CO < BaCl_2$
B) $H_2 < Ne < CO < HF < BaCl_2$
C) $H_2 < HF < Ne < CO < BaCl_2$
D) $H_2 < Ne < CO < BaCl_2 < HF$
E) $BaCl_2 < H_2 < Ne < CO < HF$

18. **(5 points)** Two isomers of *p*-nitrophenol and *o*-nitrophenol are shown below.



p-nitrophenol



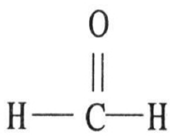
o-nitrophenol

(a) One isomer has a melting point of 44 °C and the other 144 °C. Which isomer has which melting point? **(b)** Explain why.

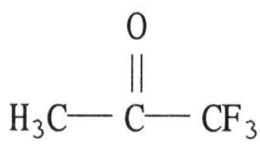
去年有一道小题考了同一个分子式但不同结构式对于分子间作用力大小的影响

5. Which one of the following substances will have hydrogen bonding as one of its intermolecular forces?

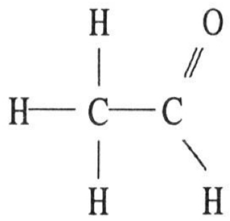
A)



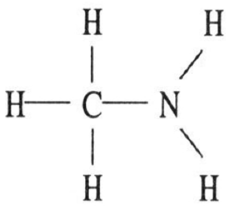
B)



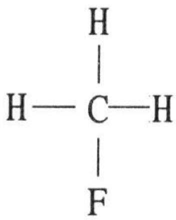
C)



D)



E)

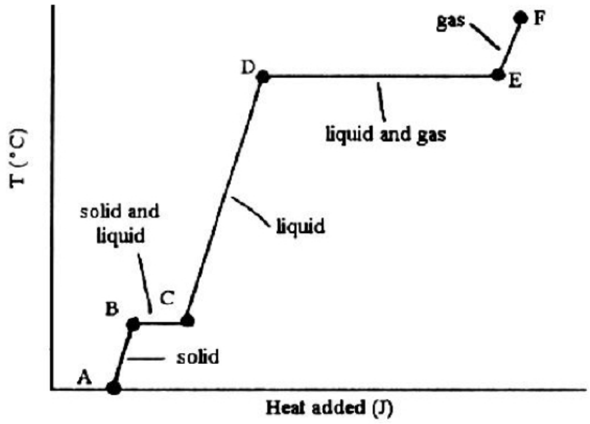


9. Large intermolecular forces in a substance are shown by ________.
A) low vapor pressure
B) high boiling point
C) high heats of fusion and vaporization
D) high critical temperatures and pressures
E) all of the above

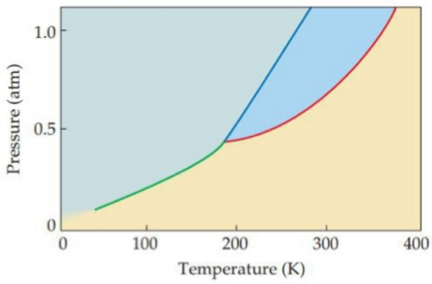
Point 2 相图相关

11. The phase changes B → C and D → E in the heating curve shown below are not associated with temperature increases because the heat energy is used up to ______.

A) increase distances between molecules
B) break intramolecular bonds
C) rearrange atoms within molecules
D) increase the velocity of molecules
E) increase the density of the sample



15. Based on the phase diagram of a hypothetical substance below, what happens to it as it is heated from 50 K to 200 K at 0.25 atm?



A) It melts at about 125 K.
B) It sublimes at about 125 K.
C) It condenses at about 125 K.
D) It boils at about 125 K.
E) It reaches the triple point at about 200 K.

注意哪个地方是supercritical fluid（去年考过）

Point 3 蒸汽压

13. In general, the vapor pressure of a substance increases as __________ increases.

A) viscosity
B) temperature
C) hydrogen bonding
D) molecular weight
E) surface tension

12. The vapor pressure of any substance at its normal boiling point is ________.

A) 1 Pa
B) 1 torr
C) 1 atm
D) equal to atmospheric pressure
E) equal to the vapor pressure of water

注意蒸汽压和气压的区别（一个是针对溶液，一个是针对气体）

蒸汽压更像是一种产生气体的能力

Point 4 液晶

14. Molecules with only single bonds do not generally exhibit liquid-crystalline properties because ___________.
A) molecules with only single bonds are too big to exhibit liquid-crystalline properties
B) molecules without multiple bonds lack the flexibility necessary for alignment
C) molecules with single bonds are gases
D) molecules without multiple bonds lack the rigidity necessary for alignment
E) molecules without multiple bonds are too small to exhibit liquid-crystalline properties

Ch 13

Point 1 溶剂溶质间的相互作用

2. Indicate the main type of solute–solvent interaction HCl in acetonitrile (CH_3CN).
A) ion-ion forces B) ion-dipole C) London dispersion D) dipole-dipole
E) hydrogen bonding

6. The process of a substance sticking to the surface of another is called __________.
A) absorption
B) diffusion
C) effusion
D) adsorption
E) coagulation

1. Which of the following explanations accounts for the fact that the ion-solvent interaction is greater for Li^+ than for K^+?
A) Li^+ is of lower mass than K^+.
B) The ionization energy of Li is higher than that for K.
C) Li^+ has a smaller ionic radius than K^+.
D) Li has a lower density than K.
E) Li reacts with water more slowly than K.

Point 2 溶解度（饱和）相关问题

8. Which of the following substances is more likely to dissolve in H_2O?
A) CCl_4
B) Ar
C) C_6H_6
D) CH_3OH

10. Indicate which statement is false in the formation of a solution.
A) A solute will dissolve in a solvent if solute–solute interactions are smaller than solute-solvent interactions.
B) An increase in entropy favors mixing.
C) In making a solution, the enthalpy of mixing is always a positive number
D) NaCl dissolves in water but not in benzene because the water–ion interactions are stronger than benzene–ion interactions.

永远是放热

11. The solubility of $MnSO_4$ monohydrate in water at 20 °C is 70.0 g per 100.0 mL of water. A solution at 20 °C that is 4.22 *M* in $MnSO_4$ monohydrate is best described as a(n) ________ solution. The formula weight of $MnSO_4$ monohydrate is 168.99 g/mol.
A) hydrated
B) solvated
C) saturated
D) supersaturated
E) unsaturated

6. A 5.25% by mass aqueous solution of NaI can be prepared by dissolving ________ .
A) 5.25 mol NaI in 1.00 L solution
B) 1.00 g NaI in 5.25 g water
C) 5.25 g NaI in 94.75 mL water
D) 5.25 mol NaI in 1.00 kg water
E) 5.25 g NaI in 100 g water

12. A solution contains 22 ppm of benzene. The density of the solution is 1.00 g/mL. This means that ________.
A) there are 22 mg of benzene in 1.0 g of this solution
B) 100 g of the solution contains 22 g of benzene
C) 1.0 g of the solution contains 22×10^{-6} g of benzene
D) 1.0 L of the solution contains 22 g of benzene
E) the solution is 22% by mass of benzene

16. Calculate the molality of a 10.0% (by mass) aqueous solution of hydrochloric acid.
A) 0.274 *m*
B) 2.74 *m*
C) 3.04 *m*
D) 4.33 *m*
E) The density of the solution is needed to solve the problem.

注意到计算过程中各种各样单位的表示

Point 3 溶液的依数性

3. Of the following, a 0.1 *M* aqueous solution of ________ will have the lowest freezing point.
A) NaCl
B) $Al(NO_3)_3$
C) K_2CrO_4
D) Na_2SO_4
E) sucrose

等于在问particles的浓度

7. Seawater contains 3.4 g of salts for every liter of solution. Assuming that the solute consists entirely of NaCl (in fact, over 90% of the salt is indeed NaCl), calculate the osmotic pressure of seawater at 20 ℃. (R = 0.0821 L·atm·K^{-1}·mol^{-1})
A) 272 atm B) 2.8 atm C) 2.69 atm D) 20.4 atm E) 817 atm

5. The vapor pressure of benzene, C_6H_6, is 100.0 torr at 26.1 ℃. Assuming Raoult's law is obeyed, how many moles of a nonvolatile solute must be added to 100.0 mL of benzene to decrease its vapor pressure by 10.0% at 26.1 ℃? The density of benzene is 0.8765 g/cm^3.
A) 0.011237 B) 0.11237 C) 0.1248 D) 0.1282 E) 8.765

20. (**a**) Does a 0.10 *m* aqueous solution of NaCl have a higher boiling point, a lower boiling point, or the same boiling point as a 0.10 *m* aqueous solution of $C_6H_{12}O_6$? Explain. (**b**) The experimental boiling point of the NaCl solution is lower than that calculated assuming that NaCl is completely dissociated in solution. Why is this the case?

(**a**) A 0.10 *m* aqueous solution of NaCl have a **higher** boiling point.
Boiling-point elevation is directly related to total moles of dissolved particles. NaCl is a strong electrolyte, The 0.10 *m* NaCl produces twice as many dissolved particles as 0.10 *m* $C_6H_{12}O_6$. So its boiling point is higher than that of 0.10 *m* $C_6H_{12}O_6$.

(**b**)
In solutions of strong electrolytes like NaCl, **ion pairing** reduces the effective number of particles in solution, decreasing the change in boiling point. The actual boiling point is then lower than the calculated boiling point for a 0.10 *m* solution.

21. List the following aqueous solutions in order of increasing boiling point: 0.120 *m* glucose, 0.050 *m* LiBr, 0.050 *m* $Zn(NO_3)_2$.

该类型为期末必考的简答题排序

Point 4 胶体相关问题

13. Which of the following <u>cannot</u> be a colloid?
A) an emulsion
B) an aerosol
C) a homogeneous mixture
D) a foam
E) All of the above are colloids.

胶体不是溶液

15. An "emulsifying agent" is a compound that helps stabilize a hydrophobic colloid in a hydrophilic solvent (or a hydrophilic colloid in a hydrophobic solvent). Which of the following choices is the best emulsifying agent?
A) CH_3COOH
B) $CH_3CH_2CH_2COOH$
C) $CH_3(CH_2)_{11}COOH$
D) $CH_3(CH_2)_{11}COONa$

Ch 14

Point 1 反应速率的影响因素和单位

1. A burning splint will burn more vigorously in pure oxygen than in air because ________.
A) oxygen is a reactant in combustion and concentration of oxygen is higher in pure oxygen than is in air
B) oxygen is a catalyst for combustion
C) oxygen is a product of combustion
D) nitrogen is a product of combustion and the system reaches equilibrium at a lower temperature
E) nitrogen is a reactant in combustion and its low concentration in pure oxygen catalyzes the combustion

2. Of the following, all are valid units for a reaction rate except ________.
A) mol/L B) *M*/s C) mol/hr D) g/s E) mol/L-hr

3. Consider the following reaction:
$$3A \rightarrow 2B$$
The average rate of appearance of B is given by $\Delta[B]/\Delta t$. Comparing the rate of appearance of B and the rate of disappearance of A, we get $\Delta[B]/\Delta t =$ _______ $\times (-\Delta[A]/\Delta t)$.
A) -2/3 B) +2/3 C) -3/2 D) +1 E) +3/2

注意正负号

Point 2 反应级数（以下几乎为必考选择题）

4. The data in the table below were obtained for the reaction:
A + B → C

Experiment Number	[A] (*M*)	[B] (*M*)	Initial Rate (*M*/s)
1	0.451	0.885	1.13
2	0.451	1.77	1.13
3	1.35	0.885	10.17

The rate law for this reaction is rate = ____________.
A) $k[A]^2[B]$ B) $k[A][B]$ C) $k[A]^2$ D) $k[A]^2[B]^2$ E) $k[C]$

5. The elementary reaction $2\ NO_2(g) \rightarrow 2\ NO(g) + O_2(g)$
is second order in NO_2 and the rate constant at 600 K is $6.77 \times 10^{-1}\ M^{-1} \cdot s^{-1}$.
The reaction half-life at this temperature when $[NO_2]_0 = 0.45\ M$ is __________ s. 3.5
A) 3.3 B) 1.0 C) 0.98 D) 1.5 E) 0.30

6. The rate constant of a first-order process that has a half-life of 3.50 min is ________ s^{-1}
A) 0.693 B) 3.30×10^{-2} C) 1.98 D) 0.198 E) 3.30×10^{-3}

3. Which of the following statements is true concerning the reaction given below?
$2\ H_2S(g) + O_2(g) \rightarrow 2\ S(s) + 2\ H_2O(g)$
A) The rate law is Rate = $k[H_2S]^2[O_2]$.
B) The reaction is second-order in $H_2S(g)$ and first-order in $O_2(g)$.
C) The reaction is first-order in $H_2S(g)$ and second-order in $O_2(g)$.
D) The rate law is Rate = $k[H_2S][O_2]$.
E) The rate law may be determined only by experiment.

Point 3 rate law factor

8. The rate of a reaction depends on __________.
A) collision frequency
B) collision energy
C) collision orientation
D) all of the above
E) none of the above

12. What is the magnitude of k for a first-order reaction at 60.00 °C if Ea = 121 kJ/mol and rate constant is $1.75 \times 10^{-1}\ s^{-1}$ at 20.00 °C?
A) 385 B) 136 C) 68 D) 34 E) 17

14. The reaction below is first order in $[H_2O_2]$:

$$2\ H_2O_2(l) \rightarrow 2\ H_2O(l) + O_2(g)$$

A solution originally at 0.600 M H_2O_2 is found to be 0.075 M after 54 min. The half-life for this reaction is ________ min.
A) 6.8   B) 18   C) 14   D) 28   E) 54

如果没有那么恰好，常规该怎么做？

Point 4 基元反应及反应机理

11. A possible mechanism for the gas phase reaction of NO and H_2 is as follows:

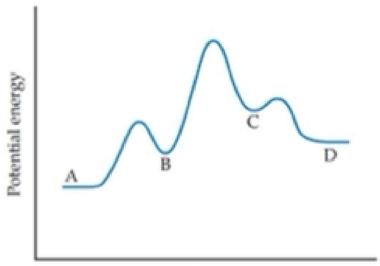
Step 1 $\quad 2\ NO \xrightarrow{k_1} N_2O_2$

Step 2 $\quad N_2O_2 + H_2 \xrightarrow{k_2} N_2O + H_2O$

Step 3 $\quad N_2O + H_2 \xrightarrow{k_3} N_2 + H_2O$

Which of the following statements concerning this mechanism is not directly supported by the information provided?
A) Step 1 is the rate determining step.
B) N_2O_2 is an intermediate.
C) There is no catalyst in this reaction.
D) The rate law for step 1 is rate = $k_1[NO]^2$.
E) All steps are bimolecular reactions.

Part B Short questions (Write legibly and show all work for all steps in the problem)
16. (5 points) Consider the following energy profile in the reaction A → D.



(a) How many elementary reactions are in the reaction mechanism? **(b)** How many intermediates are formed in the reaction A → D? And indicate them. **(c)** Which step is rate determining? **(d)** For the overall reaction, is ΔE positive, negative, or zero?

7. Below is a proposed mechanism for the decomposition of H_2O_2.

$H_2O_2 + I^- \rightarrow H_2O + IO^-$ slow

$H_2O_2 + IO^- \rightarrow H_2O + O_2 + I^-$ fast

Which of the following statements is incorrect?

A) IO^- is a catalyst.

B) I^- is a catalyst.

C) The net reaction is $2H_2O_2 \rightarrow 2H_2O + O_2$.

D) The reaction is first-order with respect to $[I^-]$.

E) The reaction is first-order with respect to $[H_2O_2]$.

slow/fast的这种如果没有写step1，step2，默认上面的方程式是step1！！！

18. Which of the following reactions is not an example of homogeneous catalysis?

A) $2O_3(g) \xrightarrow{Cl\cdot(g)} 3O_2(g)$

B) $S_2O_8^{2-}(aq) + 2I^-(aq) \xrightarrow{Fe^{2+}(aq)} 2SO_4^{2-}(aq) + I_2(aq)$

C) $2H_2O_2(aq) \xrightarrow{Br^-(aq)} 2H_2O(l) + O_2(g)$

D) $N_2(g) + 3H_2(g) \xrightarrow{Fe(s)} 2NH_3(g)$

E) $CH_3CO_2CH_3(aq) + H_2O(l) \xrightarrow{H^+(aq)} CH_3CO_2H(aq) + CH_3OH(aq)$

此为催化剂容易忽略的一点

20. (8 points) The reaction between ethyl iodide (C_2H_5I) and hydroxide ion in ethanol (C_2H_5OH) solution, has an activation energy of 86.8 kJ/mol and a frequency factor (A) of $2.10 \times 10^{11}\ M^{-1}s^{-1}$.

$$C_2H_5I\ (alcohol) + OH^-\ (alcohol) \rightarrow C_2H_5OH\ (l) + I^-\ (alcohol)$$

(a) Predict the rate constant for the reaction at 40 °C.
(b) A solution of KOH in ethanol is made up by dissolving 0.343 g KOH in ethanol to form 279.0 mL of solution. Similarly, 1.496 g of C_2H_5I is dissolved in ethanol to form 279.0 mL of solution. Equal volumes of the two solutions are mixed. Assuming the reaction is first order in each reactant, what is the initial rate at 40 °C?
(c) Assuming the frequency factor and activation energy do not change as a function of temperature, calculate the rate constant for the reaction at 50 °C.

Ch 15

Point 1 平衡常数及相关计算

5. The equilibrium-constant expression depends on the ________ of the reaction.
A) stoichiometry and mechanism
B) mechanism
C) stoichiometry
D) the quantities of reactants and products initially present
E) temperature

14. At 500 K the equilibrium constant for the reaction $2NO(g) + Cl_2(g) \rightleftharpoons 2NOCl(g)$ is $K_p = 52.0$. An equilibrium mixture of the three gases has partial pressures of 0.0950 atm and 0.171 atm for NO and Cl_2, respectively. What is the partial pressure of NOCl in the mixture?

A) 0.845 atm B) 2.97×10^{-5} atm C) 0.283 atm D) 8.02×10^{-2} atm E) 5.45×10^{-3} atm

16. Consider the equilibrium that is established in a saturated solution of silver chloride, $Ag^+(aq) + Cl^-(aq) \rightleftharpoons AgCl(s)$. If NaCl solution is added to this solution, what will happen to the concentration of Ag^+ and Cl^- ions in solution?

A) $[Ag^+]$ and $[Cl^-]$ will both increase
B) $[Ag^+]$ and $[Cl^-]$ will both decrease
C) $[Ag^+]$ will increase and $[Cl^-]$ will decrease
D) $[Ag^+]$ will decrease and $[Cl^-]$ will increase
E) neither $[Ag^+]$ nor $[Cl^-]$ will change.

Part B short questions--Write legibly and show all work for all steps in the problem. For mathematical relationships show both the equations used and substitution of values into the equation. Maintain correct units throughout your calculations and report your answers with the correct significant figures.（14 points）

17. Consider the following equilibrium, for which $\Delta H < 0$,
$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$.
Of the following changes, predict the equilibrium for the reaction above will shift to right, left or no change.
(**a**) $O_2(g)$ is added to the system.
(**b**) the reaction mixture is heated.
(**c**) the volume of the reaction vessel is doubled.
(**d**) a catalyst is added to the mixture.
(**e**) the total pressure of the system is increased by adding a noble gas.
(**f**) $SO_3(g)$ is removed from the system.

18. The following equilibria were attained at 823 K:
$CoO(s) + H_2(g) \rightleftharpoons Co(s) + H_2O(g)$ $K_c = 67$ K_{c1}
$CoO(s) + CO(g) \rightleftharpoons Co(s) + CO_2(g)$ $K_c = 490$ K_{c2}
Based on these equilibria, calculate the equilibrium constant for $H_2(g) + CO_2(g) \rightleftharpoons CO(g) + H_2O(g)$ at 823 K.

注意Kp和Kc之间的转换关系，见下面这道题

18. Phosphorus trichloride gas and chlorine gas react to form phosphorus pentachloride gas:

$PCl_3(g) + Cl_2(g) \rightleftharpoons PCl_5(g)$.

A 7.5-L gas vessel is charged with a mixture of $PCl_3(g)$ and $Cl_2(g)$, which is allowed to equilibrate at 450. K. At equilibrium the partial pressures of the three gases are $P[PCl_3(g)] = 0.124$ at m, $P[Cl_2(g)] = 0.157$ atm, and $P[PCl_5(g)] = 1.30$ atm. **(a)** What is the value of K_p at this temperature? **(b)** Does the equilibrium favor reactants or products? **(c)** Calculate K_c for this reaction at 450. K.

3. For the reaction

$2NO_2(g) \rightleftharpoons N_2O_4(g)$

K_p = 8.8 when pressures are measured in atmospheres. Under which of the following conditions will the reaction proceed in the forward direction?

	$NO_2(g)$	$N_2O_4(g)$
A)	0.200 atm	0.352 atm
B)	250 mmHg	400 mmHg
C)	0.00255 atm	0.000134 atm
D)	46.5 mmHg	82.3 mmHg
E)	0.138 atm	0.764 atm

Point 2 Haber process

2. Which one of the following is true concerning the Haber process?

A) It is a process used for shifting equilibrium positions to the right for more economical chemical synthesis of a variety of substances.

B) It is a process used for the synthesis of ammonia.

C) It is another way of stating Le Châtelier's principle.

D) It is an industrial synthesis of sodium chloride that was discovered by Karl Haber.

E) It is a process for the synthesis of elemental chlorine.